

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Seventh Semester of B. Tech. Examination (C.E. / I.T.)

Nov-2013

CE403 Data Warehousing & Data Mining (DW&DM)

Date: 19.11.2013, Tuesday Time: 10:00 a.m. To 01:00 p.m.

Maximum Marks: 70

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Rough work is to be done in the last page of main supplementary, please don't write anything on the question paper.
5. Indicate clearly, the option(s) you attempt along with its respective question no.

SECTION-I

Q-1 Answer the following questions.

- a. Describe the steps involved in data mining when viewed as a process of Knowledge Discovery. 3
- b. Describe why an online transaction processing system (OLTP) is not suitable for analytical reporting 3
- c. Explain why businesses are driven to employ data warehouse technology. Describe the role of business intelligence in today's marketplace. 3
- d. How Data Mining differ from common SQL processing? Explain by example. 2

Q-2

[A] The Apriori algorithm makes use of prior knowledge of subset support properties. Prove that all nonempty subsets of a frequent itemsets must also be frequent. 4

[B] In data warehouse technology, a multiple dimensional view can be implemented by a relational database technique (ROLAP), or by a multidimensional database technique (MOLAP), or by a hybrid database technique (HOLAP), Briefly describe each implementation technique. 4

OR

[B] In real-world data tuples with missing values for some attributes are a common occurrence. Describe various methods for handling this problem. 4

[C] State why for the integration of multiple heterogeneous information sources, many companies in industry prefer the update-driven approach (which constructs and uses data warehouses,) rather than the query-driven approach (which applies wrappers and integrators). Describe situations where the query-driven approach is preferable over the update-driven approach. 4

OR

[C] Data cleaning routines can be used to fill in missing values, smooth noisy data, identify outliers and correct the data inconsistency. Explain how to deal with each of them in terms of problems, in brief. Why data cleansing and data transformation function is considered to be a vital task in the integration process. What do you mean by Data Reduction and what are the different processes of data reduction? 4

Q-3

- [A] Give the significance and importance of the Terms: Mean, Variance, Standard deviation and central tendency. 4

OR

- [A] What do you mean by association rule, for what purpose it is being used, explain with example. How do we find interestingness of association rules? 4
- [B] Define *lift*. What is the relation between *support*, *confidence* and *lift* for $X \rightarrow Y$? 4

OR

- [B] Which are the requirements of a good clustering algorithm? 4
- [C] Compare the advantages and disadvantages of eager classification (e.g., decision tree, Bayesian, neural network) versus lazy classification (e.g., k-nearest neighbor, case-based reasoning). 4

OR

- [C] What are services provided by each of 3-tier in 3-tier architecture of warehouse. What type of back up facility is available with warehouse of your choice? 4

SECTION-II

Q-4

1. What is attribute relevance analysis? Where is it used? Describe entropy-based method to compute information-gain. 3
2. How to find the task relevant data from the domain of interest? 4
3. What are the advantages and disadvantages of decision tree induction approach for the classification over other approaches of classification? 4

Q-5

- [A] For the number series given below: 4
18,22,25,42,28,43,33,35,56,28.
1. Cal. Mean absolute deviation.
2. Cal. Std. deviation.
- [B] When is data mining appropriate for a Data warehouse? Justify with reasons and examples. 4
- [C] Where are supervised and unsupervised learning used and what purpose do they serve? Is classification a supervised or an unsupervised form of learning? What do you mean by accuracy of a classifier? 4

OR

Q-5

- [A] Describe **k-means** clustering algorithm in terms of the following criteria : (1) shapes of clusters that can be determined ; (2) input parameters that must be specified ; and (3) limitations (4) scalability. 4
- [B] What data mining functions does the business need? Can they be performed alternatively by data query processing or statistical analysis? 4
- [C] Justify the use of Bayes theorem in Bayesian Classification. Give an Example. 4

Q-6

- [A] For class characterization, what are the major differences between a data cube based implementation and a relational implementation such as AOI? Discuss which method is the efficient and under which condition? 4

[B] Consider the following transaction database:

4

TID	Item List
1	1,2,5
2	2,4
3	2,3
4	1,2,4
5	1,3
6	2,3
7	1,3
8	1,2,3,5
9	1,2,4

- Find the frequent pattern using Apriori algorithm.

[C] What are the challenges in the various fields associated with Data Mining? 4
How data mining can be useful for education and financial institutions?

OR

[C] Consider the following data.

4

	B	~B
A	200	45
~A	50	150

Find *correlation* between A and B.